

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2024

Bachelor in Computer Applications

Course Title: **Data Structures & Algorithms**

Code No: **CACS201** Semester: **III**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. Differentiate between stack and queue. What are the general applications of a stack?
3. What is a linked list? How doubly linked list is different from circular linked list? Explain with example.
4. What is recursion and recursive function? Write a recursive function to compute Fibonacci number.
5. How does collision occur during hashing? Explain any two hashing functions.
6. What is an AVL tree? Create an AVL tree from the following data: 18, 12, 14, 8, 5, 25, 31, 24, 27
7. What are deterministic and non-deterministic algorithms? Explain the use of Big Oh notation to measure the complexity of an algorithm with an example.
8. Implement the quick sort to sort the following data items: 12, 1, 14, 7, 2, 10, 4, 7, 22, 6, 15

Group C

Attempt any TWO questions [2x10=20]

9. What are the differences between linear queue and circular queue? Write an algorithm to enqueue and dequeue data elements in a circular queue.
10. What is B-tree? How insertions and deletions of elements can be done in a B-tree.
11. Explain the different ways to represent a graph. For the following graph use Prim's algorithm to find a minimum spanning tree starting from the node 'A'.